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PII: S1000-9345(26)00047-7

DOI: https://doi.org/10.1016/j.cjme.2026.100258

Reference: CJME100258

To appear in: *Chinese Journal of Mechanical Engineering*

Received date: 3 January 2025
Revised date: 8 December 2025
Accepted date: 13 February 2026

Please cite this article as: Minglong Guo, Fukuan Tian, Zhaocheng Wei, Xiangzhou Ding and Shengxian Liu, A new tool deformation compensation model considering the coupling of machining geometry and cutting force, *Chinese Journal of Mechanical Engineering,* (2026) doi:https://doi.org/10.1016/j.cjme.2026.100258

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A new tool deformation compensation model considering the coupling of machining geometry and cutting force

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Abstract
Ruled surface machining is achieved through continuous moving cutting with a large axial depth, and the tool deformation error is inevitable. In this paper, a tool compensation model considering the coupling of cutting force and machining geometry is proposed for the five-axis side milling of a ruled surface with a flat end mill. This model focuses on the coupling relationship between cutting force, cutting geometry, and tool deformation. Firstly, the three analytical sub-algorithms are established, and the cyclic iterative method is used to solve the equilibrium state of tool deformation. Further, with the coincidence of the design surface and the machining surface as the constraint condition, the tool axis vector and radial cutting depth correction are gradually searched and calculated to achieve tool deformation compensation. The experimental results show that the maximum error after compensation is 0.0543 mm, the maximum error without compensation is 0.2853 mm, and the machining accuracy is improved by about 80.97%.

Keywords: Tool deformation, Tool compensation, Ruled surface, Five axis machining, Cutting force

1 Introduction

The parts characterized by the ruled surface are widely used in aviation, petroleum, chemical and other fields, and their machining accuracy directly determines the performance of equipment. Taking the typical ruled blade surface of an integral shaft disc impeller as an example, the machining process is constrained by the geometry of the flow channel, and the tool system has a large overhang. And the machining of the ruled surface adopts a one-time forming process with large axial cutting depth, resulting in weak stiffness of the tool system and easy tool deformation. Especially in the finishing stage, the problem of tool deformation is the main reason for the substandard machining accuracy of parts [1, 2]. Precision machining at the micro-nano level is one of the important trends for the future development of the manufacturing industry [3]. For this reason, the current common methods involve reciprocating cutting or reducing the feed rate. These measures can to some extent alleviate the tool deformation, but greatly sacrifice the machining efficiency. Therefore, in-depth exploration of the mechanism of tool deformation in the machining process of ruled surfaces, and adopting reasonable compensation measures is an effective method to actively reduce tool deformation errors. At present, many scholars have conducted extensive research on tool deformation and error compensation, which has played a positive role in improving the surface accuracy.

Research on tool deformation usually equates the tool to a cantilever beam, and calculates the deformation error based on the cutting force load. Kline et al. [4] first studied the influence of static deformation of flat end mill on surface geometry, and proposed a cantilever beam structure that can predict surface geometric errors. Budak et al. [5] studied the deformation error of flat end mill. The cutting force was discretized along the cutting edge, and a cantilever beam model with equivalent discrete force was proposed. Its advantage is that there is no need to calculate the application point of total cutting force, and there is no theoretical error. Lim et al. [6] established a stepped cantilever beam model with equivalent concentrated force by considering the difference in stiffness between the tool handle and tool body of ball end mill. Wei et al. [7] established an integral stepped cantilever beam model for tool deformation during static machining of ball end mill. The bending stiffness coefficient of the tool was gradually identified through the overhang experiment method. Aiming at the deflection deformation of the micro ball end mill in micro-milling, Yuan et al. [8] obtained the cutting force through numerical integration, which provided a theoretical basis for machining error control and manufacturing process optimization. Considering that the stiffness of thin-walled parts varies with material removal and the tool contact position, Li et al. [9] regarded the flat end mill as a cantilever beam structure and proposed a method to predict the surface shape errors caused by flexible workpiece and tool deformation. The prediction accuracy of this method has been verified in the five-axis impeller milling experiment. The work is verified by the experiment of five-axis side cutting of impeller. Wang et al. [10] obtained an analytical solution of flat end mill deformation through the cantilever beam theory. By distributing flexible cutting force on the workpiece model, they proposed a flexible calculation method of deflection error. Based on the principle of minimum total potential energy and Timoshenko beam theory, Liu et al. [11] established a mathematical model considering assembly

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error and variable section geometry of micro-tool to achieve the prediction of radial oscillation. Considering the size effect and minimum chip thickness in micro-milling, Zhang et al. [12] established a cutting force model under the combined action of plow force and shear force. They regarded cutting force as a distributed load acting on the cutting edge, and achieved the prediction model of tool deformation. The establishment of a cantilever beam model requires accurate tool dimensions, and some scholars have also achieved accurate prediction of tool deformation through finite element analysis and optimization algorithms. Bhattacharya et al. [13] used the modeling methods of finite element and bond graph to predict the three-dimensional tool elastic deformation and workpiece contour error during flat end mill machining. The experimental results show that the contour error mainly depends on the radial and tangential deflection of the tool. In order to reduce the deformation caused by cutting force during the machining of thin-walled parts such as blades, Cai et al. [14] proposed a tool orientation optimization algorithm based on stiffness matching, which can effectively reduce machining deformation. In order to achieve better accuracy control in five-axis surface milling, Duan et al. [15] established a directional optimization model for tool deformation to suppress the mean value and volatility of deformation errors. Considering the influence of tool deformation on tool path and instantaneous undeformed chip thickness, Moges et al. [16] proposed an algorithm to modify tool center position and tool path, and established a model of cutting force and surface error in micro-milling. For the free-form surface machining of AL7075 material, Ozdemir et al. [17] determined the influence of tool materials and machining parameters on tool deformation and proposed a mathematical model. Experiments showed that image processing could detect tool deviation faster and more simply. Based on the general multi-parameter model and sub-component method of tool structure, Fan et al. [18] proposed a calculation method for the bending deformation of milling tools with complex structures.

In terms of tool deformation error compensation, scholars usually use techniques such as analysis, measurement, and mathematical modeling to artificially introduce an additional error source into the machining system, so that they and deformation error cancel each other. Based on the cutting force model proposed by Kline, Depince et al. [19] analyzed the influence of flexible tool machining deformation on workpiece dimensional errors, and proposed an improved error mirror compensation method. In order to improve the machining and detection accuracy, Cho et al. [20] used a closed-loop structure to compensate for the geometric and measurement errors of the three-axis CNC machining center, and corrected the tool path according to the error distribution on the basis of the RBF network model. Ma et al. [21] used the measurement data to alter the CAD model of the machined surface, so as to re-correct the tool path and compensate for the tool deformation error on the machining surface. In the process of side milling, considering radial dimension error, installation error, tool deformation and wear factors, Yu et al. [22] proposed an integrated tool contour error recognition, machining accuracy prediction and machining error compensation method. Combined with machining parameters such as tool geometry, initial surface and tool path, Zeroudi et al. [23] proposed a tool deformation calculation and compensation method based on free-form surface cutting force prediction model. This method can predict surface shape and roughness directly based on CAM data, which opens the way for complete virtual milling methods. Heng et al. [24] proposed a real-time tool offset compensation technology to improve the accuracy of micro-milling. The tool position change and axial error caused by tool deformation were estimated by cutting force and compensated by applying an additional equivalent feed. Tang et al. [25] considered the influence of surface curvature on deflection when designing geometric adaptive control compensation system. By compensating for tool deformation errors in surface geometry end face milling, they solved the problem of tool deformation caused by workpiece curvature. Yin et al. [26] established an overall stiffness model for the robot spiral arm and further fitted the relationship between machining error and cutting depth. A discrete control system for the error compensation of stone carving robotic arm is constructed using multiple iterations. In order to improve the machining accuracy and efficiency during the milling of thin-walled parts, Chen et al. [27] successfully established a modified tool path to compensate volume errors according to the deformation data of the tool position file. Gui et al. [28] established a tool workpiece contact model by combining discrete thinking with analytical geometry. Based on the simulation prediction of five-axis milling force, the machining error of thin-walled parts is compensated by optimizing the tool axis vector. By integrating robot motion data, processing data, and cutting force signals, Gu et al. [29] proposed a data-based adaptive cutting parameter adjustment strategy, which can be applied to weakly rigid tool systems.

The above research on tool deformation and deformation compensation is of great positive significance for improving surface machining accuracy. However, research on the five-axis machining of curved surfaces needs to be further deepened. During the deformation process of weakly-stiffness cutting tool, the tool deformation, cutting force, and cutting geometry have a coupling relationship, and there exists an equilibrium state between them, which is the key to improving the accuracy of the tool deformation compensation. At the same time, the introduction of compensated machining parameters will break the existing equilibrium state, and further change the tool deformation error, requiring a recalculation of the compensated machining parameters.

Therefore, this paper focuses on the influence mechanism of machining geometry and cutting force on tool deformation

and tool compensation, and proposes a more accurate five-axis machining error compensation model for ruled surfaces. The main content includes three parts: feedback and coupling relationship, core sub-algorithm, and deformation compensation process. Among them, the sub-algorithm includes three parts: cutting geometry, milling force modeling, and tool deformation error. Finally, experiments and simulations are also arranged to demonstrate the tool compensation model.

2 Feedback and Coupling in Compensation

As shown in Fig. 1, taking the typical five-axis side milling of the ruled blade surface as an example, in order to ensure machining efficiency and surface quality, the flat end mill is fed along the blade length direction, and the cutter axis vector is parallel to the surface generatrix. In this condition, the cutting force is large, the tool overhang stiffness is weak, which easily leads to tool deformation in the plane perpendicular to the tool axis. Taking the instantaneous tool position point as the analysis object, the tool is subjected to cutting forces in three directions simultaneously, causing deformation along the workpiece normal direction $\boldsymbol{n}_w$ and away from the feed direction $\boldsymbol{f}$. Among them, the material under-cutting caused by the deformation δ_f will be removed in the subsequent cutting process, and the deformation δ_n will directly affect the machining accuracy, which is the most important sensitive error source that should be concerned. Meanwhile, another deformation characteristic is increasing along the axis, mainly because the local stiffness decreases with the overhang length.

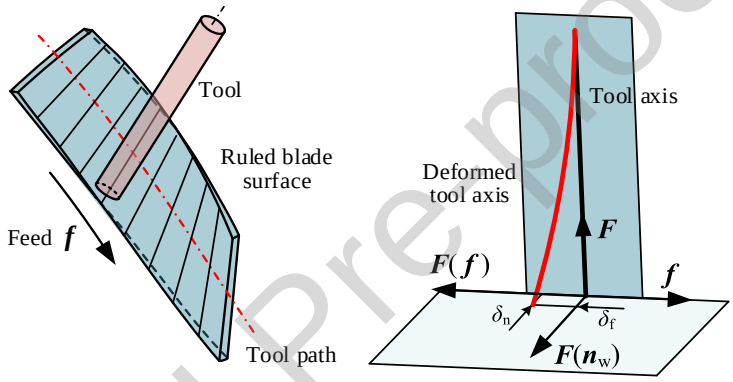

Fig. 1. Deformation of ruled surface with large axis depth

Machining geometry refers to the spatial positional relationship between the tool and the workpiece, which is mainly influenced by parameters such as the tool axis and cutting depth. The characterization methods include the tool contact area and the cutting edge immersion interval. The former refers to the area where the tool rotation contour overlaps with the workpiece, while the latter refers to the edge segments that actually participate in material cutting. As shown in Fig. 2, there are two closed feedback paths in the tool compensation system. In path I, the tool contact affects the cutting, cutting force affects tool deformation error, and the tool deformation in turn affects the tool contact by changing the preset position of the tool relative to the workpiece. The machining parameters, tool contact, cutting force, tool deformation, and radial cutting depth correction of path II also form a closed-loop feedback system. The machining parameters affect the tool deformation through the tool contact, thereby changing the tool axis and radial depth correction, and the changes of the two in turn update the input values of the machining parameters.

In summary, the key to tool compensation is to solve the equilibrium state of path I and the ideal surface state of path II, for which the cutting geometry, cutting force, and tool deformation error are the core.

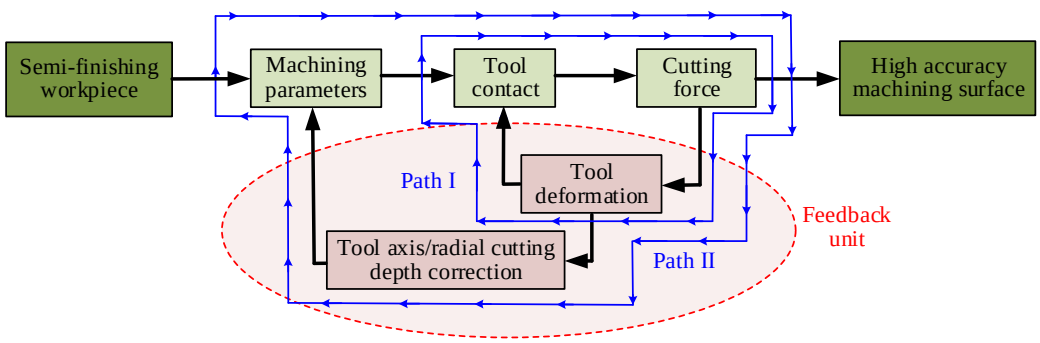

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Fig. 2. Analysis of cutting force and machining geometry

3 Core Sub-algorithm of Tool Deformation

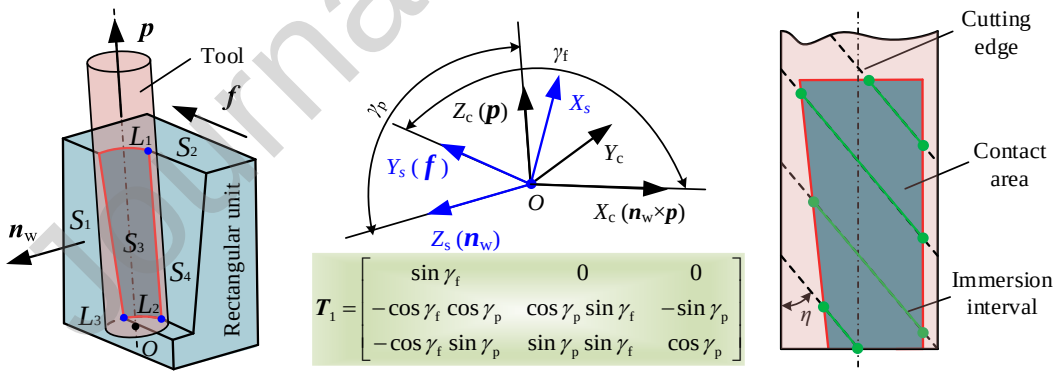
In five-axis machining of curved surfaces, the tool path is a spatial curve, and the spatial relationship between the tool and the surface is arbitrary. The contour of a ruled blade belongs to a small curvature surface. Based on the early work of our research group, the idea of differential discretization can be applied to tool error prediction and compensation modeling. Considering the characteristic that the tool path of the machine tool in machining is formed by linear interpolation, each tool position point can be approximated as a five-axis machining on an inclined plane. Considering that the tool path of machine tool machining is formed by linear interpolation, each tool point is approximated as a five-axis oblique-plane oblique plane machining. The line connecting two consecutive tool points is the feed direction of the current tool point. So, the basic research units for cutting geometry, cutting force, and tool deformation have become simple inclined plane five-axis machining with stable cutting conditions [30-32].

3.1 Cutting Geometry

The position and motion relationship between the tool and the small rectangular unit have arbitrariness, and the contact area and immersion interval are related to the radial cutting depth d_n, axial cutting depth d_p, feed direction $\boldsymbol{f}$, tool axis vector $\boldsymbol{p}$, and tool geometry, as shown in Fig. 3(a). Accurately obtaining the cutting edge immersion interval is a prerequisite for cutting force modeling.

The coordinate point of the i-th cutting edge curve of the flat end mill ($x_{c,i}$ $y_{c,i}$ $z_{c,i}$) is expressed as Eq. (1) in the tool coordinate system O-$X_cY_cZ_c$. Among them, R is the radius, η is the helix angle, h_r is the effective cutting height of the tool, θ_0 is the edge rotation angle, defined clockwise with the positive direction of the axis Y_c. As shown in Fig. 3(b), origin O is the center of the tool end face. The Z_c axis along the tool axis $\boldsymbol{p}$, X_c axis is defined as the tool axis $\boldsymbol{p}$ cross rectangle normal $\boldsymbol{n}_w$, the right-hand rule determines the Y_c axis.

$$\begin{cases} x_{c,i} = R\sin(\theta_{0,i} - z_{c,i}\tan\eta R^{-1}), \\ y_{c,i} = R\cos(\theta_{0,i} - z_{c,i}\tan\eta R^{-1}), \end{cases} \quad 0 \le z_{c,i} \le h_r. \quad (1)$$



$$\boldsymbol{T}_1 = \begin{bmatrix} \sin\gamma_f & 0 & 0 \\ -\cos\gamma_f\cos\gamma_p & \cos\gamma_p\sin\gamma_f & -\sin\gamma_p \\ -\cos\gamma_f\sin\gamma_p & \sin\gamma_p\sin\gamma_f & \cos\gamma_p \end{bmatrix}$$

(a) Relationship between tool and rectangular

(b) Space coordinate system

(c) Cutting geometry expansion

Fig. 3. Analysis of cutting geometry in five-axis side milling

As shown in Fig. 3(a), the cutting geometry expansion between the tool and the workpiece is different from traditional three-axis machining. The cutting criterion based on simple window function is no longer applicable. Therefore, the participation of edge micro-elements in material cutting can be solved using the spatial enclosure method, which needs to simultaneously satisfy the following three positional relationships. The first is the lower side of the radial unmachined surface S_1, the second is the lower side of the axial unmachined surface S_2, and the third is the intersection line between the machining surface S_3 and the already machined surface S_4, which is the feed side of the spatial curve $L_1L_2L_3$. In order to better describe the constraint conditions, the second coordinate system O-$X_sY_sZ_s$ is defined, as shown in Fig. 3(b). The normal $\boldsymbol{n}_w$ is the Z_s axis, the feed direction $\boldsymbol{f}$ is the Y_s axis, and the right-hand rule determines the X_s axis. The mathematical expression of the

above conditions is shown in Eq. (2), where γ_p represents the angle between the vectors $\boldsymbol{p}$ and normal $\boldsymbol{n}_w$, and γ_f represents the angle between the vector $\boldsymbol{f}$ and X_c axis.

$$\begin{cases} z_{s,i} \le d_n - R\sin\gamma_p, \\ x_{s,i} \le d_p, \\ y_{s,i} \le y_{s\min}. \end{cases} \tag{2}$$

Based on the Lagrangian multiplier method to solve Eq. (2) for y_{smin}, the point coordinates (x_l y_l z_l) on the spatial curve $L_1L_2L_3$ satisfy the objective function in O-$X_cY_cZ_c$:

$$\min\ g(x_l\ y_l\ z_l) = -y_l\sin\gamma_p + z_l\cos\gamma_p. \tag{3}$$

The constraint condition of Eq. (3) is that the curve point is located on the tool rotation contour, as shown in Eq. (4). Furthermore, solve the coordinate points (x_l y_l z_l) for each value of x_l, and multiply the matrix $\boldsymbol{T}_1$ to transform them into the second coordinate system (x_s y_{smin} z_{smin}).

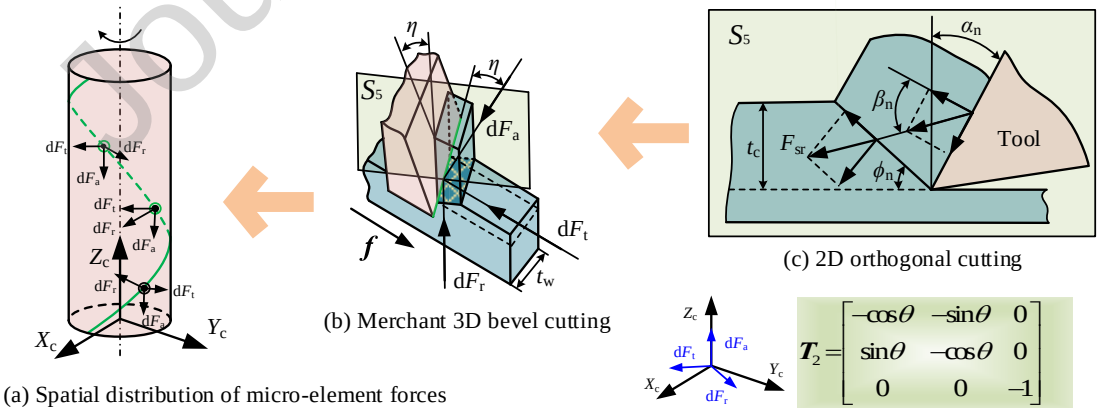
$$x_l^2 + y_l^2 = R^2. \tag{4}$$

At this point, the search algorithm is used to judge all micro-elements along the edge, extract the immersion interval [ζ_d ζ_u] involved in material cutting, and finally approach the exact solution step by step through the golden section algorithm or dichotomy. ζ is defined as the angle between the tool axis and the line connecting the micro-element point and the cutting height end face center, which facilitates a unified and monotonic description of the micro-element points on the side and bottom edges, as shown in Eq. (5).

$$\zeta = \arctan[R(h_r - z_c)^{-1}]. \tag{5}$$

3.2 Cutting Force Modeling

Force is an important fundamental physical quantity in the cutting process [33, 34], and tool deformation is one of its adverse consequences. As shown in Fig. 4(a), the basic idea for solving the cutting force is the principle of calculus. Firstly, the cutting edge is discretized into micro-elements, and their cutting state and force are analyzed [33]. Then, sum up the three-dimensional force vectors of the micro-elements on the immersion interval, and finally obtain the overall cutting force of the milling tool at any rotation time. Considering the angle η between the cutting speed and the cutting edge direction, while ignoring the interaction between the micro-elements, the cutting edge micro-elements are regarded as independent straight edge bevel cutting units.



(a) Spatial distribution of micro-element forces

(b) Merchant 3D bevel cutting

(c) 2D orthogonal cutting

$$\boldsymbol{T}_2 = \begin{bmatrix} -\cos\theta & -\sin\theta & 0 \\ \sin\theta & -\cos\theta & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Fig. 4. Analysis of cutting force on flat end mill

As shown in Fig. 4(c), on the section S_5 perpendicular to the cutting edge, based on the plastic shear and chip flow theory of the cutting layer material, the resultant force on the shear plane is equal to that on the rake face. The shear resultant force F_{sr} is expressed as Eq. (6), where β_n is the normal friction angle, α_n is the normal rake angle, and ϕ_n is the shear angle. The

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relationship between them conforms to the minimum energy theory proposed by Melcante. τ_s represents the yield shear stress, which is a function $f(\varepsilon, \acute{\varepsilon}, T)$ of strain ε, strain rate $\acute{\varepsilon}$, and temperature T.

$$F_{sr} = \frac{\tau_s t_w t_c}{\sin\phi_n \cos(\phi_n + \beta_n - \alpha_n)}, \tag{6}$$

$$4\phi_n = \pi - 2\beta_n + 2\alpha_n. \tag{7}$$

In Eq. (6), $t_c(\theta, \zeta)$ is the cutting layer thickness, defined as the projection of feed vector $\boldsymbol{f}=[0\ f_v\ 0]$ on the normal $\boldsymbol{n}_c$ of tool rotation profile, denoted as $\boldsymbol{f}\boldsymbol{T}_1^{-1}\boldsymbol{n}_c$. Among them, $\boldsymbol{n}_c=[x_c\ y_c\ 0]$. $t_w(\zeta)$ is the cutting layer width, denoted as dz_c.

As shown in Fig. 4(b), combined with the relationship between force and velocity of orthogonal cutting and bevel cutting, and considering that the edge inclination angle and chip flow angle are approximately the helix angle η, the tangential, radial, and axial force components of micro-elements on the 3D bevel cutting section can be expressed as Eq. (8). Among them, $\theta=\theta_0+R^{-1}z_c\tan\eta$.

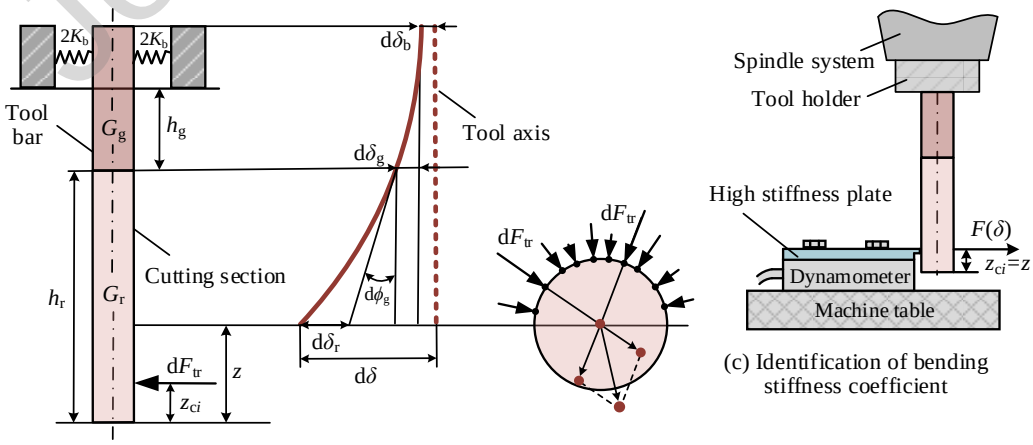
$$\begin{bmatrix} dF_{t,i}(\theta,\zeta) \\ dF_{r,i}(\theta,\zeta) \\ dF_{a,i}(\theta,\zeta) \end{bmatrix} = \frac{F_{sr}\sin\phi_n}{\sqrt{\sin^2\phi_n + \tan^2\eta\sin^2\beta_n}} \begin{bmatrix} \sin(2\phi_n) + \tan^2\eta\sin\beta_n \\ \cos(2\phi_n)\sec\eta \\ \tan\eta[\sin(2\phi_n) - \sin\beta_n] \end{bmatrix}. \tag{8}$$

The magnitude and direction of the cutting force of each micro-element are different, depending on their spatial position and cutting parameters. Multiply the micro-element force vector $[F_t\ F_r\ F_a]^T$ by $\boldsymbol{T}_2$ and convert it to tool coordinate system $O\text{-}X_cY_cZ_c$. Perform vector summation of the series of micro-element forces within the immersion interval $[\zeta_d\ \zeta_u]$ and N tool teeth. The overall cutting force is expressed as follows:

$$[F_{xc}(\theta_0)\quad F_{yc}(\theta_0)\quad F_{zc}(\theta_0)]^T = \sum_{i=1}^{N}\int_{\zeta_d}^{\zeta_u}\left\{[dF_{t,i}(\theta,\zeta)\quad dF_{r,i}(\theta,\zeta)\quad dF_{a,i}(\theta,\zeta)]^T\boldsymbol{T}_2\right\}. \tag{9}$$

3.3 Tool Deformation Error

The cutting force load is the fundamental cause of tool deformation, and its characteristics of large axis depth side milling are that the action area is large and distributed in multiple spiral tool immersion intervals. Considering the small elastic deformation of the tool, the milling force load is discretized. Through independent analysis of the tool deformation under the action of cutting micro-forces, and ignoring the interference between micro-elements, the total deformation is obtained by vector summation of all deformation elements. As the cutting tool rotates, the cutting force shows a fluctuating trend. Considering the high-frequency fluctuation characteristics, the transient cutting force corresponding to the maximum tool deformation error is extracted as the analysis object.



(a) Analysis of tool deformation (b) Vector sum of deformation (c) Identification of bending stiffness coefficient

Fig. 5. Deformation analysis of flat end mill

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As shown in Fig. 5(a), the large overhanging tool system is equivalent to a cantilever beam structure, and the total deformation is the comprehensive result of tool holder, tool cutting section, tool bar rotation, and tool holder. Considering the large axial stiffness of the tool, the influence of micro axial force $\mathrm{d}F_a$ is ignored. The tangential force $\mathrm{d}F_t$ acts on the sidewall of the tool, ignoring its effect on the torsional deformation. In the plane perpendicular to the tool axis, the effective deformation force is determined by the vector sum of tangential force $\mathrm{d}F_t$ and radial force $\mathrm{d}F_r$, as shown in Fig. 5(b). The deformation $\mathrm{d}\delta$ generated by any cutting edge micro-element at a distance of z from the tool end face is expressed as follows:

$$\mathrm{d}\delta_i(\theta_0,\zeta,z)=\mathrm{d}\delta_{\mathrm{g},i}(\theta_0,\zeta)+\mathrm{d}\delta_{\mathrm{r},i}(\theta_0,\zeta)+\mathrm{d}\varphi_{\mathrm{g},i}(\theta_0,\zeta)+\mathrm{d}\delta_{\mathrm{b},i}(\theta_0,\zeta). \tag{10}$$

In Eq. (10), $\mathrm{d}\delta_g$ represents the deflection deformation of the tool bar, $\mathrm{d}\delta_r$ represents the deflection deformation of the tool cutting section, $\mathrm{d}\varphi_g$ represents the bending angle deformation of the tool bar, and $\mathrm{d}\delta_b$ represents the deformation of the tool holder. The calculation expression is shown in Eq. (11) [35]. Among them, $\mathrm{d}F_{\mathrm{tr},i}(\theta_0, \zeta)$ is the vector sum of the $\mathrm{d}F_t$ and $\mathrm{d}F_r$, h_g is the overhang length of the tool bar, G_g and G_r are the bending stiffness coefficients of the tool bar and the tool cutting, and K_b is the stiffness of the tool holder.

$$\begin{cases}\mathrm{d}\delta_{\mathrm{g},i}(\theta_0,\zeta,z)=\mathrm{d}F_{\mathrm{tr},i}(\theta_0,\zeta)[2h_{\mathrm{g}}{}^3+3h_{\mathrm{g}}{}^2(h_{\mathrm{r}}-z_{ci})](6G_{\mathrm{g}})^{-1},\\ \mathrm{d}\delta_{\mathrm{r},i}(\theta_0,\zeta,z)=\mathrm{d}F_{\mathrm{tr},i}(\theta_0,\zeta)[<z_{ci}-z>^3-(h_{\mathrm{r}}-z)^3+3(h_{\mathrm{r}}-z)^2(h_{\mathrm{r}}-z_{ci})](6G_{\mathrm{r}})^{-1},\\ \mathrm{d}\varphi_{\mathrm{g},i}(\theta_0,\zeta,z)=\mathrm{d}F_{\mathrm{tr},i}(\theta_0,\zeta)[h_{\mathrm{g}}{}^2+2h_{\mathrm{g}}(h_{\mathrm{r}}-z_{ci})](h_{\mathrm{r}}-z)(2G_{\mathrm{g}})^{-1},\\ \mathrm{d}\delta_{\mathrm{b},i}(\theta_0,\zeta,z)=\mathrm{d}F_{\mathrm{tr},i}(\theta_0,\zeta)K_{\mathrm{b}}{}^{-1}.\end{cases} \tag{11}$$

Based on Eqs. (10) and (8), traverse and calculate the deformation $\mathrm{d}\delta_i(\theta_0, \zeta, z)$ of all micro-element forces. Considering the spatial characteristics of deformation direction consistent with the force load $\mathrm{d}F_{\mathrm{tr},i}(\theta_0, \zeta)$, the deformation amount $\delta(\theta_0, z)$ at height z under the comprehensive action of all micro-element forces is obtained by vector summation, as shown in Eq. (12):

$$\delta(\theta_0,z)=\sum_{i=1}^{N}\int_{\zeta_{\mathrm{d}}}^{\zeta_{\mathrm{u}}}[\mathrm{d}\delta_i(\theta_0,\zeta,z)]. \tag{12}$$

Among them, the values of the variable z in the interval $[0\ h_g+h_r]$ can be input into Eq. (12) one by one to calculate the tool axis deviation of different heights z, and obtain the three-dimensional space curve of the deformed tool axis. This deformation can be decomposed into two parts: along the normal direction of the workpiece, $\delta_n(\theta_0, z)$, and away from the feed direction, $\delta_f(\theta_0, z)$, as shown in Fig. 1, which can be solved by projection. Each rotation angle θ_0 can calculate a set of $\delta_n(\theta_0, z)$ and $\delta_f(\theta_0, z)$. The maximum normal deformation $\delta_{\mathrm{n,max}}(z)$ within one tool rotation will be used as the object of subsequent deformation compensation.

The bending stiffness coefficient is the product of the sectional moment of inertia and the elastic modulus, and the tool bar G_g can be calculated using the equation $G_g=0.25\pi R^4E$. The geometry of the tool cutting section is complex, and this paper proposes an experimental method to identify G_r. As shown in Fig. 5(c), the high stiffness plate is fixed on the dynamometer, the plate contacts the tool edge and moves horizontally under the control of the precision machine tool, while recording displacement and dynamometer data. This experiment simulates the tool deformation error of a single edge micro-element force, and the condition $z=z_{ci}$ is used to calculate the G_r in Eq. (10). This method suggests conducting multiple experiments to improve data reliability.

4 Tool Deformation Compensation

The tool compensation strategy is to artificially introduce the machining parameter adjustment to offset the undercutting error caused by tool deformation error, ultimately making the machining geometry of the ruled surface consistent with the design geometry. The machining surface of the ruled surface at tool position point is parallel to the generatrix, and the tool deformation is essentially the result of a series of polynomial accumulations. In theory, there is no absolute ideal tool position point and tool axis that can make the tool contour sweep surface completely coincide with the designed machining surface. Considering the main influencing factors on the machining surface accuracy of the deformation component δ_n, this paper first performs linear fitting on it and compares it with the designed surface to determine the initial values for compensating the radial cutting depth d_n and angle γ_p.

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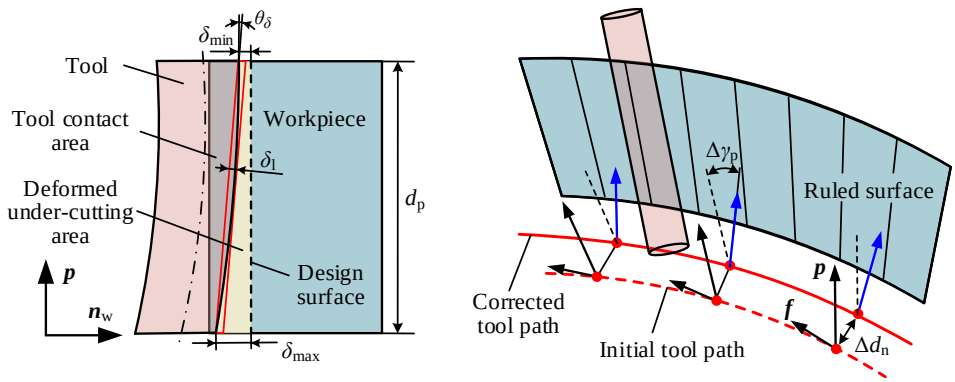

(a) Compensation of tool axis and cutting depth (b) Tool path correction

Fig. 6. Tool deformation compensation principle

This paper proposes two methods for fitting and analyzing tool deformation curve, as shown in Fig. 6(a). Based on the straightness definition, linear fitting of the deformed tool axis is carried out, and the tool compensation is shown in Eq. (13). In the formula, δ_{max} is the maximum machining error along the normal direction $\boldsymbol{n}_w$, corresponding to the deformation of the tool end face. δ_{min} is the minimum machining error, corresponding to the deformation at the top of axial cutting depth. δ_l is the straightness error of the deformed tool axis. θ_δ is the angle between the fitted line and the designed machining surface, calculated as Eq. (14).

$$\begin{cases} d_n \to d_n + \Delta d_n \ (\delta_{max} - \delta_l \cos^{-1}\theta_\delta \le \Delta d_n \le \delta_{max}), \\ \gamma_p \to \gamma_p - \Delta\gamma_p \ (\Delta\gamma_p = \theta_\delta), \end{cases} \tag{13}$$

$$\theta_\delta = \arctan[(\delta_{max} - \delta_{min}) d_p^{-1}]. \tag{14}$$

In Eq. (13), when Δd_n takes the value of $\delta_{max}-\delta_l\cos^{-1}\theta_\delta$, the tool profile is completely above the design surface, and there is no over-cutting; When the value is δ_{max}, the tool profile is completely below the design surface and there is no under-cutting.

In addition, the least squares method can be used to fit the tool axis curve with the constraint of minimum sum of squared errors. The fitted linear equation can be expressed as $y=k_p x+b_n$. At this point, the compensation for the tool angle γ_p and radial cutting depth d_n is as shown in Eq. (15). The angle compensation $\Delta\gamma_p$ is equal to the arctangent of the slope of the fitted linear equation, and the radial depth compensation Δd_n is equal to the Y-axis intercept. This compensation method will simultaneously have over-cutting and under-cutting, but the comprehensive error from the ideal contour is the smallest.

$$\begin{cases} d_n \to d_n + \Delta d_n, \quad (\Delta d_n = b_n), \\ \gamma_p \to \gamma_p + \Delta\gamma_p, \quad (\Delta\gamma_p = \arctan k_p). \end{cases} \tag{15}$$

The calculation of the tool compensation amounts Δd_n and $\Delta\gamma_p$ is a complex process, as described in the coupling feedback relationship in Section 2, which mainly includes two iterative loops of Path I and Path II. The termination condition of path I is that the cutting force, tool deformation error, and tool contact geometry reach an equilibrium state. Its quantitative expression is that the difference between two consecutive calculations of $\delta_{n,max}(z)$ is within the set tolerance range. During each iteration, the cutting edge element judgment needs to be adjusted based on the new deformation calculation results. The termination condition of path II is that the surface geometry formed by the superposition of input machining parameters and tool compensation parameters matches the design geometry. It is quantitatively expressed as the distance between the design surface and the tool profile within the set tolerance range. At each iteration, the machining parameters need to be updated based on the newly calculated radial cutting depth and tool axis vector. Moreover, each iteration in path II includes a complete cycle in path I. Based on the sub-algorithms of tool cutting geometry, cutting force, and tool deformation error established, precise tool compensation can be achieved using MATLAB numerical analysis software. The solution process is shown in Fig. 7.

In addition, important discretization principles for tool rotation motion and cutting edge curve are also used in the process, because the tool cutting geometry and cutting force in these two environments are constantly changing. The deformation information of all the edge micro-elements in these two discrete environments needs to be traversed.

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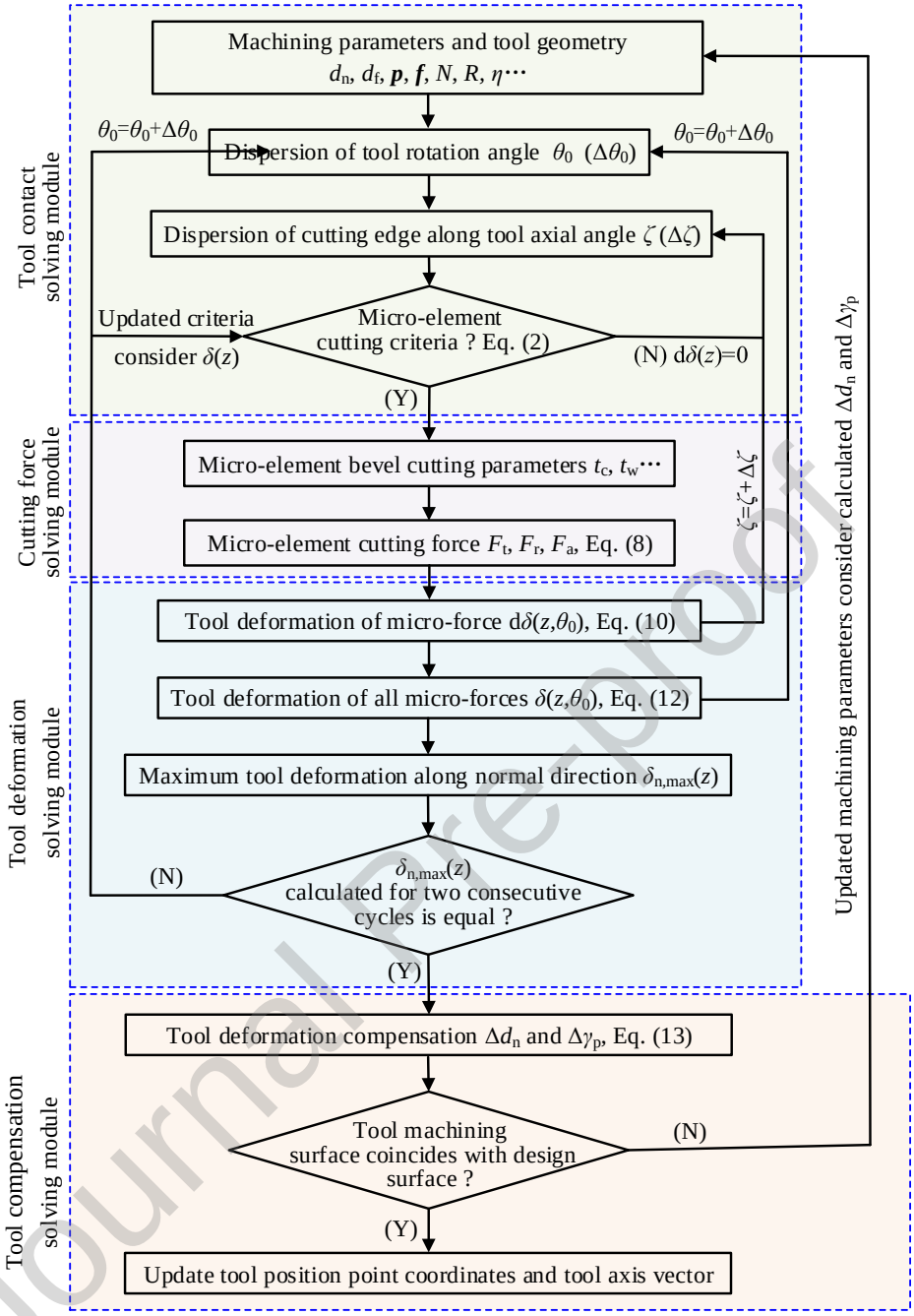


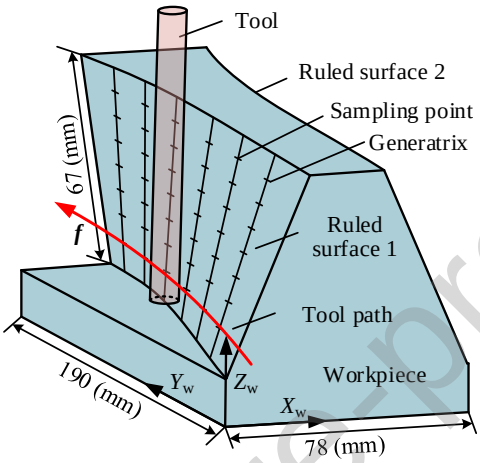
Fig. 7. Five axis side milling compensation solution process of ruled surface

The calculated compensation value can only be "written" into the tool path of finishing to obtain a high-precision machined surface. As shown in Fig. 6(b), the tool path of the five-axis side milling CNC machining of the ruled curve is linear interpolation, and the tool position point is the center of the tool end face. When deformation is not considered, the tool position point is located on the surface normal and the distance from the surface is equal to the tool radius. The tool axis vector is parallel to the generatrix. Based on the calculated Δd_n and $\Delta\gamma_p$, the updated radial cutting depth $d_n+\Delta d_n$ and tool axis angle $\gamma_p+\Delta\gamma_p$ are input into the CAM software to reproduce the tool path, which is further imported into the five-axis CNC machining center to effectively compensate for the deformation error of the tool in the normal direction of the workpiece.

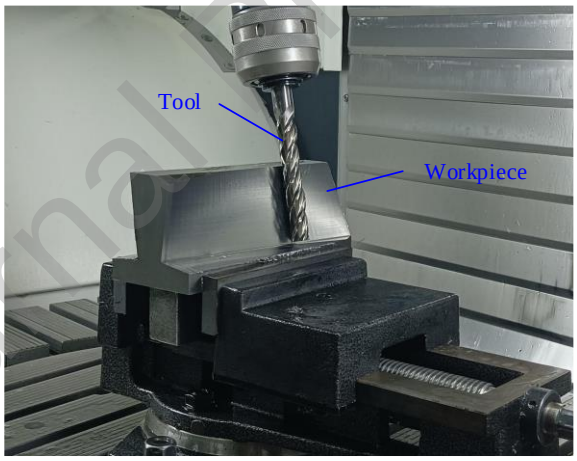
5 Experiment and Simulation

In order to verify the effectiveness of the proposed tool compensation model in this paper, the following five-axis side milling experiment of a ruled surface with a flat-end mill is arranged. The workpiece geometry is shown in Fig. 8(a). Surface

1 and Surface 2 are exactly the same, and compensated machining and non-compensated machining are carried out respectively. The generatrix of the ruled surface is 67 mm long, and the sweeping line is a free curve with a length of 191 mm. Workpiece material FV520(B), shear stress 1297.65 N/mm^2, friction coefficient 0.775. The material of the flat end mill is high-speed steel (W6Mo5Cr4V2), with an elastic modulus of 205 GPa, the diameter is 16 mm, the overhang length is 120 mm, the cutting section length is 100 mm, the helix angle is 35°, the number of teeth is 1, and the experimentally identified bending stiffness G_g=6.595×108 Nmm^2, G_r=1.311×108 Nmm^2. Down milling, while setting speed 600 r/min, feed rate 48 mm/min, theoretical design radial cutting depth 0.5 mm, axial cutting depth 60 mm, angle γ_p 90°, and angle γ_f 90°. The workpiece design and tool path generation are completed with Unigraphics NX software. The experiment equipment is DMU60monoBLOCK machining center, as shown in Fig. 8(b). The contour measurement equipment for machined surfaces is the German Zeiss PRISMO navigator three-coordinate machine, with a measurement accuracy of 0.5+L/500 μm.



(a) Workpiece geometry of ruled surface



(b) Experimental site

Fig. 8. Five axis side milling experiment of ruled surface

The cutting geometry algorithm established based on this paper can calculate the real-time changes of cutting edge immersion interval after one tool rotation cycle. As shown in Fig. 9(a), the black solid line represents the upper and lower bounds of the edge immersion interval considering tool deformation, and the blue dashed line represents the situation without deformation. Obviously, the effect of cutting force causes the tool to deform away from the workpiece, resulting in a shorter length of the edge immersion interval. The lower bound ζ_d under the two working conditions is consistent, while the upper bound ζ_u is different. The difference between the two increases with the increase of the tool rotation angle. When the rotation angle is 281°, the length of the interval considering deformation increases by 50.5%.

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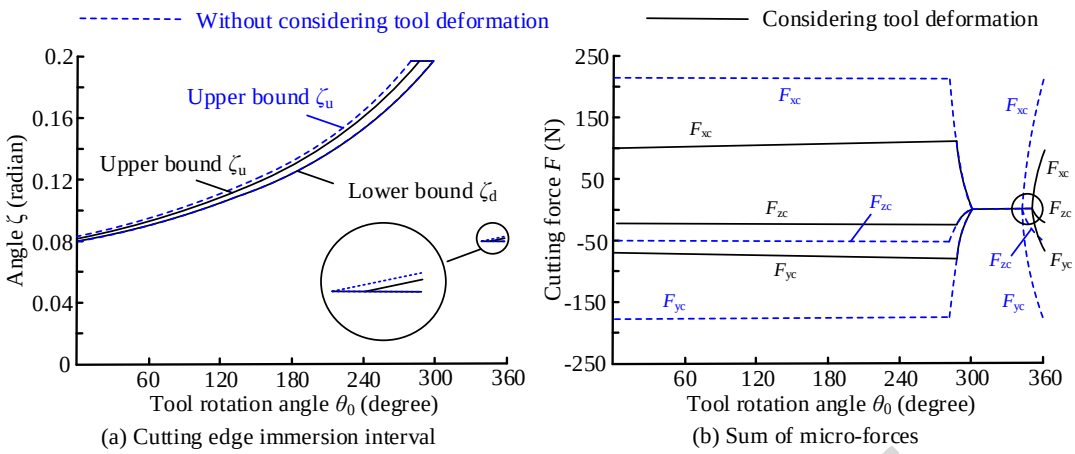


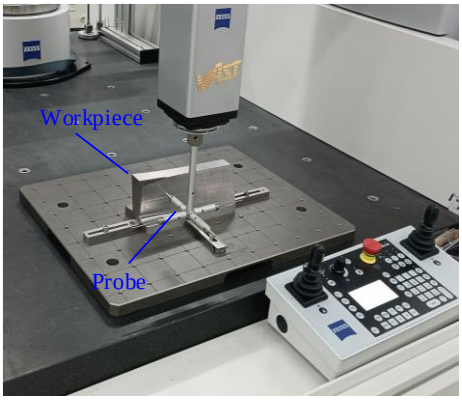
(a) Cutting edge immersion interval (b) Sum of micro-forces

Fig. 9. Comparison of edge immersion interval and cutting force

The changing trend of cutting force is synchronized with the edge immersion interval. After one cycle of tool rotation, the sum of all micro-forces vectors in the tool coordinate system is shown in Fig. 9(b). The black solid line represents the cutting force considering tool deformation, while the blue dashed line represents the case of no tool deformation. The peak values of cutting forces F_{xc}, F_{yc}, and F_{zc} with deformation are reduced by approximately 48.5%, 55.4%, and 51.4%, respectively. As indicated by the circle in Fig. 9(a), the tool deformation delays the cutting time, and the time when the cutting force occurs is also delayed, as indicated in Fig. 9(b).

Based on the tool compensation model established in this paper, the tool position point and tool axis vector for machining surface 1 have been modified. As a comparison, the machining of surface 2 was not compensated. As shown in Fig. 10(a), perform three coordinate measurements on 42×2 sampling points on the machined surface, where the Y_w and Z_w values of the sampling point coordinates are determined, and the X_w value is measured. The 42 sampling points on each machined surface are uniformly distributed along the generatrix direction and the tool path direction. In theory, the 6 generatrices in Fig. 8(a) have the same deformation in the height direction, so the 7 sampling points on the 6 generatrices are averaged to eliminate the accidental influence of measurement.

The measurement results of surface 2, as shown in point data in Fig. 10(b), indirectly reflect the actual tool deformation. The predicted tool deformation without considering the coupling relationship between machining geometry and cutting force is shown by the green line in the figure, which is much higher than the actual tool deformation, with a maximum prediction error of 0.309 mm. The tool deformation predicted based on the theoretical model in this article is shown by the red line in the figure, which is highly consistent with the experimental measurement results. The maximum prediction error is 0.022 mm, reducing the average prediction error from 122.64% to 10.88%.



(a) Measuring site

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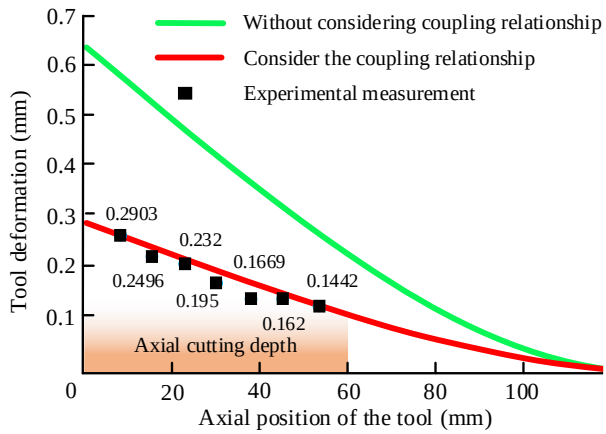

(b) Prediction and measurement of tool deformation

Fig. 10. Comparison of tool deformation for ruled surface

The measurement results of the compensated surface 1 are shown in Fig. 11, where positive values indicate under-cutting, and negative values indicate over-cutting. The maximum tool deformation error is 0.0543 mm, and the minimum value is -0.0362 mm. Surface 2 exhibits a significant under-cutting phenomenon, with a maximum tool deformation error of 0.2853 mm, and a minimum value of 0.1392 mm. This is because the tool deformation caused by cutting force shifts the cutting position, and the closer to the tool end face, the greater the machining error value. Due to the selection of the least squares method as the compensation benchmark, there are slight under-cutting and over-cutting phenomena on surface 1 after machining, which makes the machined surface closer to the design surface. The maximum tool deformation error after surface 1 compensation decreased by 80.97%, proving the effectiveness of the tool compensation model.

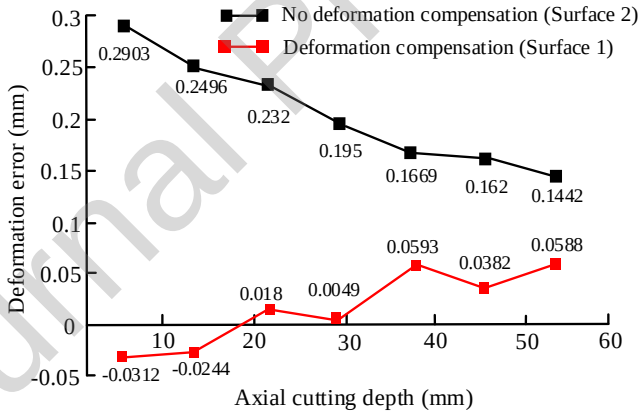

Fig. 11. Comparison of machining accuracy for ruled surface

6 Conclusions

Aiming at tool deformation in five-axis side milling of ruled surface, a tool compensation model is proposed in this paper. Specific conclusions are as follows:

(1) The coupling relationship between tool contact, cutting force, and tool deformation is discussed, while the deformation compensation mechanism among machining parameters (including tool axis and radial depth correction), tool contact, cutting force, and tool deformation is further established.

(2) The three core algorithms of cutting geometry, cutting force and tool deformation in the tool deformation model are established. The cutting-edge immersion interval is calculated by the spatial enclosure method. The cutting force calculation is based on the idea that the discrete cutting edge micro-element is an independent bevel cutting unit. The calculation principle of tool deformation is the vector sum of the deformation caused by the micro-element forces.

(3) Ultimately, a tool compensation model is proposed by correcting the tool axis vector and radial cutting depth. The arranged experiment and simulation has demonstrated the coupling relationship between tool contact geometry, cutting force, and tool deformation. The maximum machining surface error after tool compensation is 0.0362 mm, which is reduced by

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80.97%.

(4) This tool compensation model can be applied to cutting conditions combinations under different cutting parameters, flat end mill geometries, and workpiece materials. It can also be extended to machining systems with other workpiece geometries and tool types.

Funding

Supported by National Natural Science Foundation of China (Grant Nos. 52005078, 52405458 and 52075076).

Credit Authorship Contribution Statement

Minglong Guo wrote the manuscript, planning and implementing the main research work of this paper; Fukuan Tian and Zhaocheng Wei put forward the overall research idea and experimental platform; Xiangzhou Ding and Shengxian Liu assisted with English writing and verification experiments. All authors read and approved the final manuscript.

Declaration of Competing Interests

The authors declare no competing financial interests.

Acknowledgements

Not applicable.

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